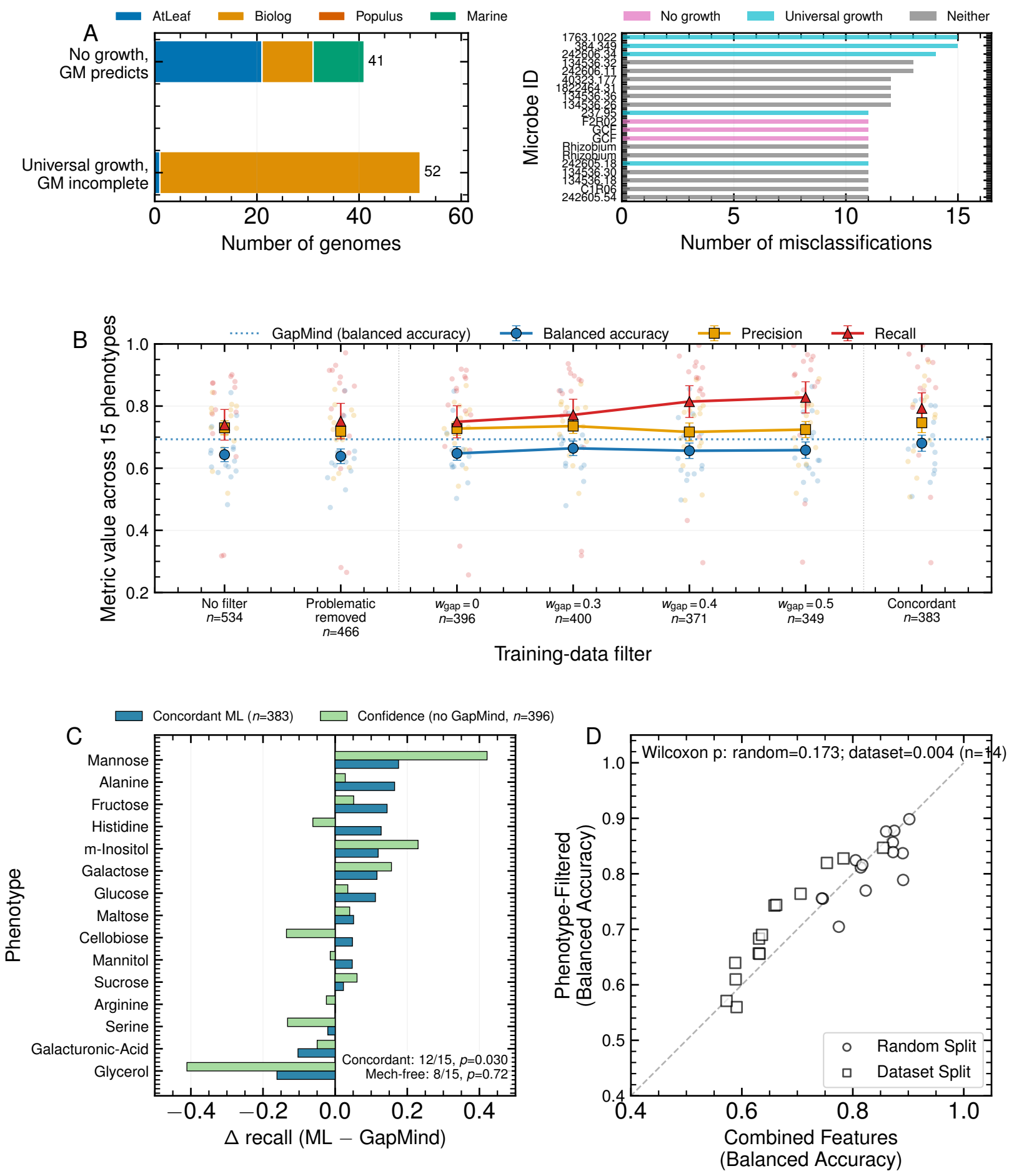
A
AtLeaf
Biolog
Populus
Marine
No growth, GM predicts
41
Universal growth, GM incomplete
52
0
20
40
60
Number of genomes
No growth
Universal growth
Neither
Microbe ID
1763.1022
384.349
242606.34
134536.32
242606.11
40323.177
1822464.31
134536.36
134536.26
237.95
F2R02
GCF
GCF
Rhizobium
Rhizobium
242605.18
134536.30
134536.18
C1R06
242605.54
0
5
10
15
Number of misclassifications
B
GapMind (balanced accuracy)
Balanced accuracy
Precision
Recall
Metric value across 15 phenotypes
1.0
0.8
0.6
0.4
0.2
No filter n=534
Problematic removed n=466
wgap = 0 n=396
wgap = 0.3 n=400
wgap = 0.4 n=371
wgap = 0.5 n=349
Concordant n=383
Training-data filter
C
Concordant ML (n=383)
Confidence (no GapMind, n=396)
Phenotype
Mannose
Alanine
Fructose
Histidine
m-Inositol
Galactose
Glucose
Maltose
Cellobiose
Mannitol
Sucrose
Arginine
Serine
Galacturonic-Acid
Glycerol
Concordant: 12/15, p=0.030
Mech-free: 8/15, p=0.72
−0.4
−0.2
0.0
0.2
0.4
Δ recall (ML − GapMind)
D
Wilcoxon p: random=0.173; dataset=0.004 (n=14)
Phenotype-Filtered (Balanced Accuracy)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Random Split
Dataset Split
0.4
0.6
0.8
1.0
Combined Features (Balanced Accuracy)